

Psychological-physical synergy of athletes based on artificial intelligence and deep learning

Received: 29 October 2025

Accepted: 23 March 2026

Published online: 25 March 2026

Cite this article as: Cheng Y., Gong T. & Zhao H. Psychological-physical synergy of athletes based on artificial intelligence and deep learning. *Sci Rep* (2026). <https://doi.org/10.1038/s41598-026-45920-4>

Yuantao Cheng, Ting Gong & Hesong Zhao

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

Psychological-Physical Synergy of Athletes Based on Artificial Intelligence and Deep Learning

Yuantao Cheng¹, Ting Gong², Hesong Zhao^{3,*}

¹Department of Physical Education, Chengdu Textile College, Chengdu, 611731, China; chengyuantao@cdtc.edu.cn

²School of Rare Metal Industries, Hejun College, Ganzhou, 342600, China; Gongting0522@163.com

³School of Humanities, Gannan Healthcare Vocational College, Ganzhou, 341000, China; zhaohs2024@163.com

*Corresponding author: Hesong Zhao, e-mail: zhaohs2024@163.com

Abstract: To address the challenge of accurately and reliably monitoring individuals' psychological states from multimodal physiological signals, this study aims to solve two core issues of existing deep learning methods: the lack of collaborative modeling and the unreliability of prediction results. This study proposes an innovative framework named Graph Attention-Bayesian Temporal Network (GABT-Net). First, the method uses Graph Attention Network (GAT) to construct each moment's multi-source physiological signals into a "physiological collaboration graph" to explicitly capture the instantaneous collaborative relationships among various indicators. Then, it introduces Bayesian Transformer and performs probabilistic prediction on the time series of collaborative states through Monte Carlo (MC) Dropout technology, thereby realizing the quantification of the model's own uncertainty. Experimental verification conducted on the public Wearable Stress and Affect Detection (WESAD) dataset shows that the GABT-Net model achieves an average accuracy of 96.2% and a macro-average F1-score of 0.9431, both of which outperform the baseline models. In terms of reliability, the Expected Calibration Error (ECE) of this model is 0.021, and the prediction accuracy on samples with the highest confidence reaches 99.10%. Ablation experiment and graph construction strategy analysis further verify the key role of GAT collaborative modeling and Bayesian time series module in performance and reliability.

Keywords: Deep learning; Graph neural network; Bayesian Transformer; Multimodal physiological signals; Pressure identification

1. Introduction

Driven by the dual demands of pursuing human extreme performance in competitive sports and safeguarding the physical and mental health of high-pressure professional groups, accurate and real-time monitoring of individuals' psychological-physiological states has become a frontier in scientific research. With the popularization of wearable sensing technology, it has become possible to continuously collect multi-dimensional physiological data such as Heart Rate Variability (HRV) and Electrodermal Activity (EDA). This opens a window for in-depth understanding of the dynamic changes in human states [1,2]. These signals are objective reflections of autonomic nervous system activity and are closely related to an individual's stress, emotional and fatigue states [3]. However, how to effectively decode deep information that reflects an individual's real state from these high-dimensional and noisy data, and overcome the limitations of traditional analysis methods in handling complex nonlinear relationships among multiple indicators, remains a challenging research topic.

Although deep learning models represented by Long Short-Term Memory (LSTM) have made significant progress in processing physiological time-series data, they still face two core bottlenecks. First, most existing models treat multi-channel signals as independent inputs or simply concatenate them, ignoring the collaborative network formed by various indicators under the regulation of the nervous system [4]. For instance, in a state of stress, the "collaborative pattern" of heart rate, respiration, and skin conductance can reveal the essence of the state more effectively than any single indicator [5]. Second, the deterministic predictions provided by existing models cannot express their own confidence levels. When an incorrect early warning may lead to major decision-making errors, the lack of quantitative evaluation of model uncertainty becomes a key obstacle preventing these technologies from being applied in real-world high-risk scenarios.

When dealing with multimodal physiological signals, existing deep learning methods often treat them as independent feature channels for simple concatenation. They ignore that the human physiological system is an organic whole, and there are specific dynamic collaborative relationships between various subsystems (such as the cardiovascular system and electrodermal activity) under different psychological states. The proposed Graph Attention-Bayesian Temporal Network (GABT-Net) is not a simple stack of existing modules. Its core lies in establishing a modeling paradigm of "physiological synergy graph". The model uses a graph attention network to reconstruct discrete physiological indicators into an instantaneous collaborative topological structure, explicitly learning the

nonlinear correlations between sympathetic and parasympathetic nerve activities at the data level. On this basis, a Bayesian Transformer is introduced to address the inherent noise interference and individual difference challenges of physiological signals. By integrating the Monte Carlo Dropout mechanism into temporal inference, the model can capture long-term dependencies and output a confidence evaluation for each prediction. This mechanism, which combines the explicit structured representation of physiological collaborative relationships with the probabilistic expression of prediction results, constitutes the essential feature that distinguishes this method from traditional models.

2. Literature Review

Currently, emotion and stress state recognition using wearable devices has become a rapidly developing research field, and the breadth of this research is evidenced by the review by Cai et al. (2023) [6]. Researchers have explored a variety of sensing modalities, ranging from traditional Electrocardiogram (ECG) and Electroencephalogram (EEG) to innovative sensing technologies. For example, Sweeney-Fanelli & Imtiaz (2024) [7] used temporal convolutional network (TCN) to analyze ECG signals, while Wang et al. (2023) [8] developed a dual-channel EEG system for emotion recognition. More cutting-edge explorations include the multifunctional nanofibers for facial expression recognition studied by Zong et al. (2025) [9]. Given the limitations of single sensors, multimodal fusion has become a mainstream strategy. Ahmed et al. (2023) [10] evaluated the effectiveness of multimodal wearable sensors in quantifying emotional states and depression. To achieve efficient fusion, researchers have proposed various methods: Nandini et al. (2025) [11] developed an integrated deep learning framework, and Bhatti et al. (2024) [12] designed an attention cross-connection fusion mechanism. Although these studies have made progress in fusing multi-source data, they generally regard the fusion process as simple aggregation or weighting at the feature level. They fail to explicitly model the inherent and dynamic collaborative relationships between different physiological signals as a structured representation, which limits the model's in-depth understanding of complex physiological states.

With the increase in model complexity, the Transformer architecture is rapidly becoming a research focus in this field due to its outstanding ability to capture long-range temporal dependencies. Shen (2025) [13] proposed TransformHR, applying the Transformer to athletes' heart rate monitoring. Toğaçar et al. (2025) [14] combined wavelet transforms with the Transformer model for detecting athletes' fatigue states. In multimodal scenarios, Phan et al. (2024) [15] designed a Transformer-driven dual-stream fusion model for real-time monitoring of leg exercises. The powerful capabilities of this architecture have also extended to biomechanical analysis, such as the BioMAT model developed by Sharifi-Renani et al. (2023) [16] for joint kinematics prediction, as well as sports action recognition [17] and athlete potential evaluation [18]. In addition, some studies have begun to explore more comprehensive technical frameworks. For example, Yu et al. (2024) [19] and Wang et al. (2022) [20] combined digital twins technology with emotion recognition, while Moser et al. (2024) [21] focused on the interpretability of deep learning models. However, although these advanced Transformer models excel in temporal modeling, they usually treat multi-channel inputs as a flattened feature sequence. They still fail to address the aforementioned collaborative modeling issue, namely, failing to effectively capture the structured relationships across different physiological modalities at each moment before conducting analysis in the temporal dimension.

Beyond model architectures, key challenges in real-world applications have also spurred new research directions. Li & Washington (2024) [22] conducted a comparative study that revealed the trade-off between personalized and generalized models, highlighting the importance of adapting to individual differences. To address the issue of sparse data annotation, Zhu et al. (2023) [23] proposed a wearable-based unsupervised network for abnormal emotion detection. Meanwhile, Liu et al. (2024) [24] explored the use of deep reinforcement learning for emotion intervention. These studies, together with the reviews by Yang et al. (2024) [25] and Wu et al. (2025) [26], have identified significant opportunities and challenges for intelligent wearable systems in healthcare and sports fields, including their potential in specific applications such as ear-worn EEG [27]. Overall, despite the substantial progress made in existing literature regarding sensor technologies, fusion methods, and temporal models, there remains a lack of a framework that can uniformly model both the instantaneous collaborative relationships (spatial structure) between physiological signals and their long-term dynamic evolution (temporal sequence). Additionally, nearly all studies focus on providing deterministic predictions, while neglecting the quantification of model uncertainty, which is crucial in high-stakes decision-making scenarios like healthcare and sports. This study aims to directly address these two core challenges by integrating GNN with Bayesian Transformer.

3. Research Model

3.1 Model Overall Architecture

The Graph Attention-Bayesian Temporal Network (GABT-Net) proposed in this study aims to construct an end-to-end framework that can uniformly model the collaborative relationships and temporal dynamics among multimodal physiological signals. Its core design idea is to decompose the complex physiological state recognition task into two key sub-problems: first, how to capture the instantaneous collaborative patterns among different physiological indicators at any moment. Second, how to learn the evolutionary laws of these collaborative patterns over time and make reliable predictions. To solve these two problems, GABT-Net organically integrates two advanced technologies: GNN [28] and Bayesian Transformer [29]. Among them, GNN is used for structured collaborative modeling of multi-modal physiological signals in the spatial dimension, where the graph attention mechanism can adaptively learn the correlation strength between different physiological indicators. Bayesian Transformer is used for probabilistic sequence modeling in the temporal dimension. By introducing Monte Carlo Dropout (MC Dropout) in the inference stage, it realizes predictive distribution estimation, thereby quantifying model uncertainty and improving result reliability. The detailed data processing flow of GABT-Net is shown in Figure 1.

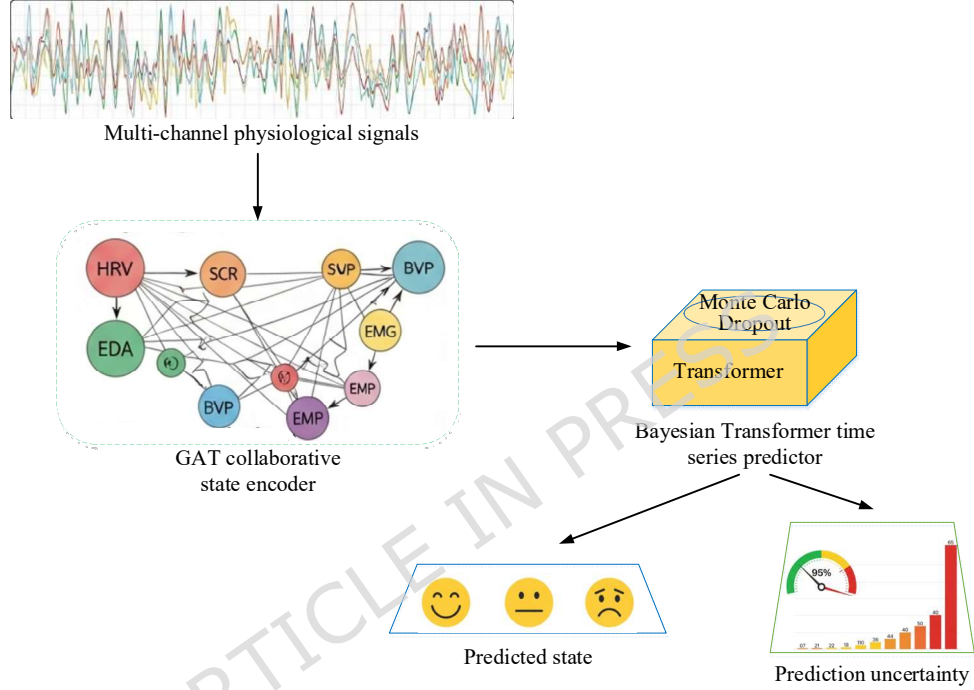


Figure. 1 Schematic diagram of GABT-Net process framework

3.2 GAT Cooperative State Coding

To transform unstructured multimodal physiological signals into relational data for model learning, a "physiological synergy diagram" is first constructed at each time window t , as shown in equation (1):

$$G_t = (V, E) \quad (1)$$

In this figure, each node $v_i \in V$ of node set V represents an independent physiological index (such as HRV, EDA, etc.), and its initial state is described by a feature vector $x_i \in R^F$, where the feature dimensions of F indicators. Given that the internal interaction mechanism of the physiological system is complex and difficult to model accurately in advance, a fully connected graph structure is adopted. That is, the edge set E contains all node pairs to avoid structural bias caused by artificial assumptions. The learning task of the correlation strength between indicators is completed adaptively by the subsequent graph attention network. Among them, GAT cooperative state coding is shown in Figure 2.

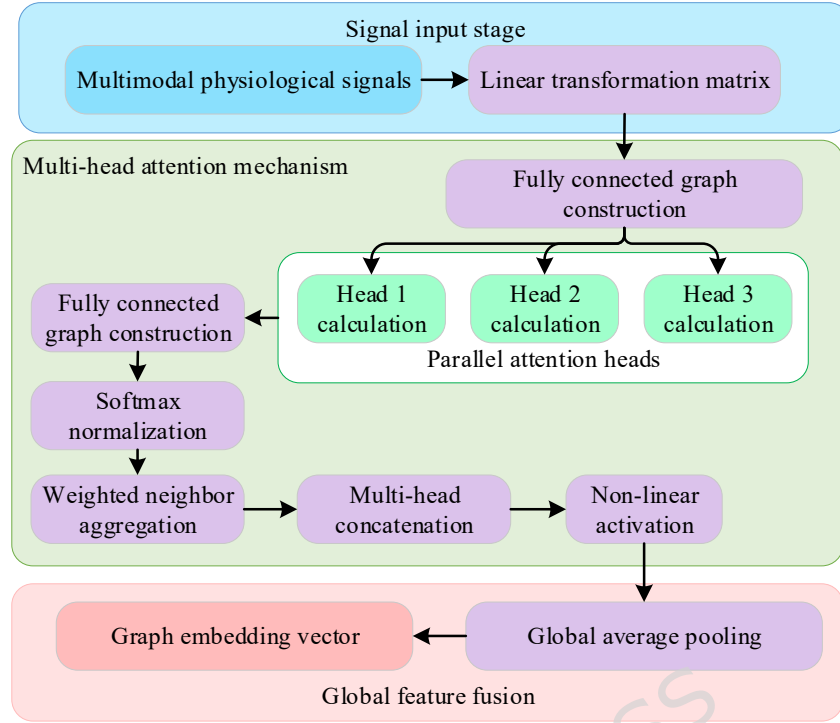


Figure. 2 Schematic diagram of GAT cooperative state encoder

Figure 2 details the internal workflow and data flow mechanism of the graph attention network-based synergy state encoder. Multimodal physiological signals are first mapped to a high-dimensional feature space via a linear transformation matrix to enhance their expressive capability. The constructed fully connected physiological graph then enters the multi-head attention mechanism module, where the model performs multiple independent attention computations in parallel to capture feature correlations across different subspaces. In the attention coefficient calculation phase, the algorithm dynamically evaluates the importance of neighboring nodes to the central node and completes weight normalization using the Softmax function. Based on these normalized weights, the model performs weighted aggregation of neighborhood information and processes it through a nonlinear activation function to capture complex physiological nonlinear relationships. After concatenation of the feature vectors output by the multi-head attention mechanism, dimensionality reduction and feature compression are conducted via a global average pooling layer, ultimately generating graph embedding vectors highly condensed with physiological synergy information. These vectors provide input representations containing rich structured information for subsequent time-series prediction.

GAT [30] is adopted to learn the deep representation of each node in the graph, and its core is to assign different weights to the neighbors of the node through the self-attention mechanism. For any node i in the graph, the model first maps the input features of all nodes to a higher dimensional space through a shared and learnable linear transformation matrix W to enhance its expressive ability. Then, for any neighbor j of node i , the model calculates the correlation strength between them through an attention function a , that is, the attention coefficient e_{ij} . In the standard GAT implementation, this function is usually a single-layer feedforward network, and its calculation process is as shown in equation (2):

$$e_{ij} = \text{LeakyReLU}([a^T(W_{x_i} \parallel W_{x_j})]) \quad (2)$$

a is a learnable weight vector, $[\cdot \parallel \cdot]$ represents vector concatenation operation, and LeakyReLU is a nonlinear activation function. e_{ij} represents the importance of the characteristics of node j to node i . To make these coefficients comparable among different neighbors, the model uses Softmax function to normalize all neighbors $j \in N_i$ of node i to get the final attention weight α_{ij} , as shown in equation (3):

$$\alpha_{ij} = \text{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in N_i} \exp(e_{ik})} \quad (3)$$

These normalized weights α_{ij} constitute a probability distribution about the neighbors of node i . With these weights, the updated feature vector h'_i of node i can be obtained by weighted summation of the transformed features of all its neighbors, and processed by a nonlinear activation function σ (such as ELU), as shown in equation (4):

$$h'_i = \sigma\left(\sum_{j \in N_i} \alpha_{ij} W_{x_j}\right) \quad (4)$$

To make the model capture rich relationship information between nodes from different angles and stabilize the training process, the Multi-Head Attention (MHA) mechanism is adopted. This mechanism performs k independent attention calculations in parallel, and uses a different set of parameters (W^k, a^k) to learn an independent attention representation each time:

$$h_i^{multi} = \parallel_{k=1}^K \sigma\left(\sum_{j \in N(i)} \alpha_{ij}^k W^k h_j\right) \quad (5)$$

$\parallel$ denotes the feature concatenation operation. The multi-head mechanism can learn multi-scale physiological collaboration patterns in different subspaces, thus enhancing the representation capability. To obtain the graph-level embedding vector that represents the overall physiological collaboration state at the current time step, a readout operation is performed on the final features of all nodes. Global average pooling is adopted:

$$H_t = \frac{1}{N} \sum_{i=1}^N h_i^{multi} \quad (6)$$

The obtained vector H_t highly compresses and integrates the collaboration relationship among all physiological indicators at the current moment, forming the input of the subsequent Bayesian time series prediction module. Compared with traditional feature concatenation or convolution modeling methods, GAT-based collaborative coding explicitly models the dependency relationships between indicators and improves the structural representation ability of mental state recognition. It adaptively learns the collaboration strength and avoids relying on unreliable prior physiological connection assumptions. This model provides interpretable attention weights, which help analyze the contribution of key physiological indicators to mental states. This module provides structured and physiologically interpretable high-quality input representations for the subsequent temporal probabilistic prediction.

3.3 Bayesian Transformer Time Series Prediction

After GAT cooperates with the state encoder to transform the multimodal physiological signal at each moment into a unified state representation vector, a time series $S = [h_{t-n}, \dots, h_t]$ which can reflect the evolution law of physiological state is obtained. To capture the complex long-term dependencies in this sequence, Transformer architecture is adopted. However, the standard Transformer provides deterministic prediction, which cannot evaluate the confidence of the model. Therefore, the core innovation of this study is to transform Transformer into approximate realization of Bayesian Neural Network (BNN), so that it can make probabilistic prediction. Through MC Dropout technology, the model is endowed with the ability to quantify its own uncertainty while maintaining strong time series modeling ability.

The standard Transformer encoder layer is the basic building block of the model. In this module, the input sequence X is linearly projected into three different spaces, and the Query, Key and Value matrices are obtained respectively, as shown in equations (7) to (9):

$$Q = XW_Q \quad (7)$$

$$K = XW_K \quad (8)$$

$$V = XW_V \quad (9)$$

W_Q, W_K, W_V are learnable weight matrices. The core idea of self-attention is to obtain the attention score of each element in the sequence to other elements by calculating the dot product of query and all keys. These scores are scaled (to prevent the gradient from being too small) and normalized by Softmax, and then used as weights to sum the value matrices. The process is defined by equation (10):

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (10)$$

d_k is the dimension of the bond vector. The MHA mechanism allows the model to pay attention to information from different locations in different representation subspaces by performing h independent attention calculations in parallel (that is, using a different set of W_Q, W_K, W_V weights for each "head"). The outputs of all the heads are spliced and projected through a linear layer again, as shown in equation (11):

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O \quad (11)$$

$\text{head}_i = \text{Attention}(XW_{Q_i}, XW_{K_i}, XW_{V_i})$. After residual connection and layer normalization, the output of the attention module is sent to a Feed-Forward Network (FFN), which performs independent nonlinear transformation on each position in the sequence, as shown in equation (12):

$$\text{FFN}(z) = \text{ReLU}(zW_1 + b_1)W_2 + b_2 \quad (12)$$

W_1, b_1, W_2, b_2 are the weights and offsets of this layer of network. At this point, a standard Transformer encoder layer has been built.

To quantify the credibility of prediction results from the perspective of probability distribution, the Dropout mechanism is kept activated during the model inference phase, and K independent random forward propagations are performed on the same input sequence. Owing to the different random neuron dropout patterns in each propagation, a distinct predicted probability distribution is generated. The final prediction result is obtained by calculating the expectation of these K random samples, as shown in equation (13):

$$\bar{p} = \frac{1}{K} \sum_{k=1}^K p_k \quad (13)$$

K represents the total number of iterations of Monte Carlo sampling (set to 100 in this study), and p_k represents the probability vector output by the k th random forward propagation. The final prediction category $\hat{y}$ of the model is the category with the largest probability value in the average probability distribution, as shown in equation (14):

$$\hat{y} = \underset{c}{\operatorname{argmax}}(\bar{p}_c) \quad (14)$$

Finally, the most important step is to quantify the uncertainty of the model. A commonly used and effective index is Predictive Entropy, which measures the confusion or uncertainty of the final average forecast distribution. Its calculation is as shown in equation (15):

$$H(\bar{p}) = -\sum_{c=1}^C \bar{p}_c \log_2(\bar{p}_c) \quad (15)$$

C is the total number of categories, and $\bar{p}_c$ is the average prediction probability of category c . A higher entropy value means that the prediction probability of the model is evenly distributed in multiple categories, indicating that the model has low confidence in this prediction and high uncertainty. On the contrary, a lower entropy value means that the model is highly confident of its prediction results. Compared with traditional deterministic time series models, the Bayesian Transformer time series prediction mechanism can retain the long-range dependency modeling ability of Transformer and realize probabilistic prediction. MC Dropout provides Bayesian approximate inference without additional parameters. It outputs interpretable uncertainty measures to improve the reliability of mental state monitoring. Among them, the pseudo-code flow of Bayesian Transformer time series prediction is shown in Table 1.

Table. 1 Pseudo-code flow chart of Bayesian Transformer time series predictor

Type	Symbol	Explanation
Input	$S \in R^{T_{seq} \times d_{model}}$	Physiological cooperative state feature sequence generated by GAT
Input	K	Number of Monte Carlo sampling iterations
Input	p	Dropout probability in Bayesian inference
Output	$\hat{y}$	Final predicted class label
Output	H	Prediction uncertainty measured by information entropy
Step	Operation description	
1	Initialize probability set: $P \leftarrow \emptyset$	
2	Begin Bayesian sequential inference via Monte Carlo sampling	
3	For $k=1$ to K do	
4	Keep Dropout activated in the Transformer encoder during inference	
5	Compute multi-head self-attention output: $H_{\text{attn}} \leftarrow \text{MultiHeadAttention}(S, S, S; \text{dropout} = p)$	
6	Pass through feedforward network: $H_{\text{enc}} \leftarrow \text{FFN}(H_{\text{attn}}; \text{dropout} = p)$	
7	Obtain classification logits: $z_k \leftarrow \text{Classifier}(H_{\text{enc}})$	

8	Convert logits to probability distribution: $P_k \leftarrow \text{Softmax}(z_k)$
9	Add current probability to set: $P \leftarrow P \cup \{P_k\}$
10	End For loop
11	Perform statistical aggregation over Monte Carlo samples
12	Compute mean predictive probability: $\bar{P} \leftarrow \frac{1}{K} \sum_{k=1}^K P_k$
13	Determine final predicted label: $\hat{y} \leftarrow \arg \max(\bar{P})$
14	Calculate prediction uncertainty (entropy): $\mathcal{H} \leftarrow - \sum_{c=1}^C \bar{P}_c \log \bar{P}_c$
15	Return $\hat{y}$ and H

4. Experimental Design and Performance Evaluation

4.1 Datasets Collection

This study uses the public WESAD dataset (<https://www.kaggle.com/datasets/orvile/wesad-wearable-stress-affect-detection-dataset>) for model training and evaluation. This dataset is a recognized benchmark in the fields of affective computing and stress detection. The dataset includes synchronous multimodal physiological signals collected from 15 subjects in three distinct states (baseline, stress, and amusement) via a chest-worn device. The signals cover ECG, EDA, respiration, body temperature, and triaxial accelerometer data. The preprocessing workflow is as follows: First, the signals are filtered and denoised, then down sampled to 64 Hz. Second, a series of well-recognized physiological state-related features are extracted from the signals of each channel, such as HRV indicators and skin conductance levels. Finally, a 60-second sliding window (with a 1-second step size) is used to segment the continuous feature sequence into samples. To ensure the generalization of model evaluation, a Leave-One-Subject-Out (LOSO) cross-validation strategy is adopted. Specifically, data from 14 subjects are used for training in turn, and data from the remaining 1 subject is used for testing.

4.2 Experimental Environment and Parameter Setting

All experiments are conducted on a server equipped with an NVIDIA RTX 4090 GPU, and the software framework is built based on Python, PyTorch, and the PyTorch Geometric library. The Adam optimizer and cross-entropy loss function are used for end-to-end training of the model. Detailed settings of the model's key hyperparameters are shown in Table 2.

Table. 2 Experimental environment configuration and hyperparameter setting table

Parameter category	Parameter name	Numerical value
Data parameter	Window Size	60 seconds
	Step Size	1 second
Training parameters	Optimizer	Adam
	Learning Rate	5e-4
	Batch Size	64
	Epochs	120
GAT module	Number of GAT layers	2
	Number of Heads	4
	Hidden layer dimension	64
Transformer module	Transformer layer number	3
	Heads of attention	4
	Feedforward network dimension	256
Bayesian module	Dropout Rate	0.3
	MC Dropout sample number (t)	100

4.3 Performance Evaluation

To comprehensively evaluate the performance of GABT-Net, a series of advanced baseline models are selected for comparison. These include Bi-LSTM and standard Transformer for temporal modeling, as well as GAT + Transformer. The classification performance of all models is measured using three core metrics: Accuracy, Macro F1-Score, and Confusion Matrix, which are presented in Figures 3-5.

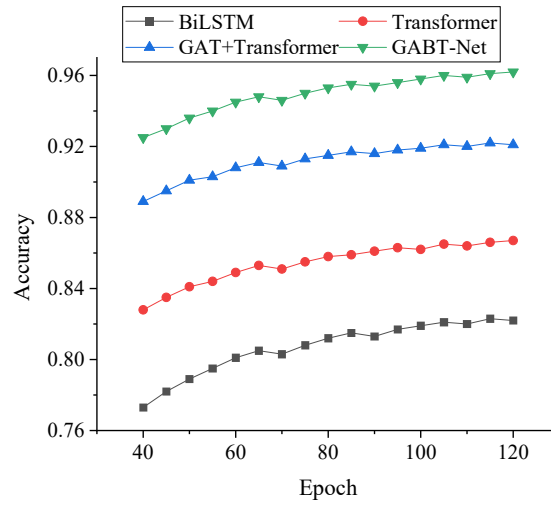


Figure. 3 Comparison of the accuracy of different algorithms

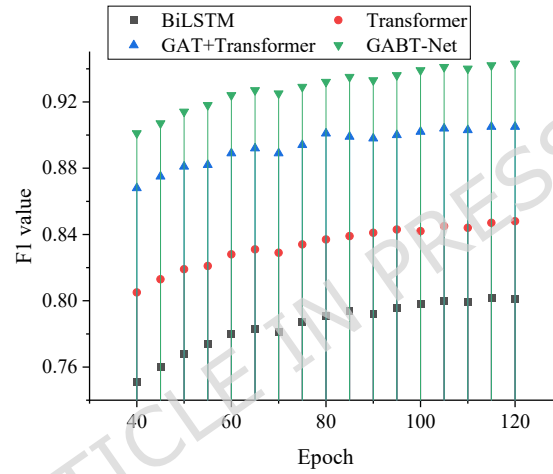


Figure. 4 Comparison of macro average F1 scores of different algorithms

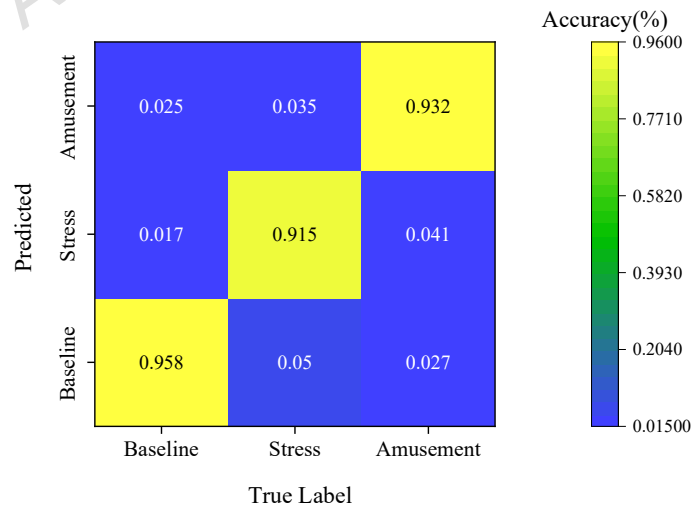


Figure. 5 Average confusion matrix diagram of GABT-Net model on LOSO cross-validation test set

Figures 3 and 4 show that GABT-Net surpasses the baseline model in accuracy and macro average F1 score during the whole training process. It finally converges to an Accuracy of 0.962 and an F1-Score of 0.9431 at the

120th epoch, which demonstrates its excellent overall performance and faster learning efficiency. To verify whether the performance improvement is statistically significant, a paired t-test was employed to compare GABT-Net with the optimal baseline model GAT+Transformer based on the Macro-F1 scores of each subject in LOSO cross-validation. The test result shows that $p < 0.05$, indicating that the improvement of GABT-Net in classification performance is statistically significant and the gain is not caused by random fluctuation. Notably, by introducing GAT to capture the collaborative relationships between physiological signals, both GAT +Transformer and GABT-Net achieve a significant performance leap compared with the standard Transformer. This highlights the importance of collaborative modeling. The confusion matrix in Figure 5 shows the classification distribution in detail. The accuracy of the model in identifying the baseline state reaches 95.80%. Confusion mainly occurs between stress and entertainment, which is consistent with the similarity of these two emotional states at physiological activation level.

To quantitatively evaluate the model's reliability, an in-depth analysis is conducted from two dimensions. Figure 6 presents the comparison results of Expected Calibration Error (ECE) between the proposed GABT-Net model and several baseline models, aiming to measure the overall accuracy of the model's prediction confidence. Figure 7 further delves into the sample level: through uncertainty-accuracy correlation analysis, it verifies whether the prediction entropy (a measure of uncertainty) output by GABT-Net can effectively indicate the correctness of a single prediction.

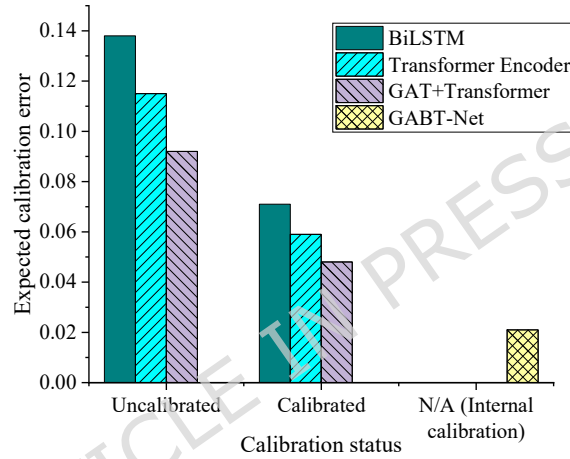


Figure. 6 Comparison chart of ECE of different models before and after calibration

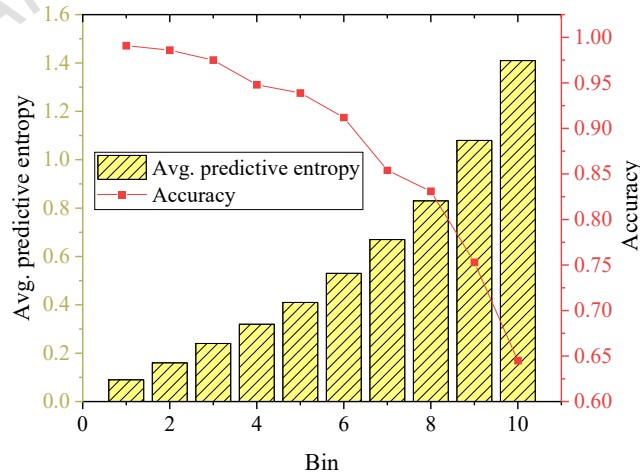


Figure. 7 Correlation analysis results of GABT-Net prediction entropy and prediction accuracy

The experimental results strongly demonstrate the significant advantages of GABT-Net in prediction reliability. In Figure 6, all deterministic baseline models exhibit high ECE values without calibration (e.g., 0.138 for Bi-LSTM), confirming the widespread overconfidence issue in deep learning models. Although post-

processing Calibrated can effectively reduce their ECE, the proposed GABT-Net achieves the optimal intrinsic calibration performance with an ECE value of 0.021, without any additional processing. Figure 7 further reveals the practical value of this reliability: there is a clear negative correlation between the model's prediction accuracy and its prediction entropy. In the sample interval where the model is most confident (Bin 1, average entropy of 0.09), the accuracy reaches as high as 99.10%. In contrast, in the interval where the model is least confident (Bin 10, average entropy of 1.41), the accuracy decreases to 64.50% accordingly. This monotonically decreasing trend—even with slight fluctuations that align with experimental randomness (e.g., Bin 8)—clearly confirms that the uncertainty output by GABT-Net is an efficient and reliable indicator. It can accurately identify predictions where the model may make errors, which is crucial for building trustworthy AI systems.

To break through the limitation of relying solely on overall accuracy, this study further evaluates the fine-grained metrics of the model under various emotional states, introducing sensitivity, specificity and F1-score, as well as calculating the Unweighted Average Recall (UAR). Table 3 presents the detailed evaluation results based on leave-one-out cross-validation.

Table. 3 Classification performance index and statistical significance test results based on one-out-of-one cross-validation

Index	Baseline state	Pressure state	Entertainment state	Overall/average
Sensitivity	0.958	0.915	0.932	0.935 (UAR)
Specificity	0.962	0.971	0.969	0.967
Accuracy rate	0.925	0.940	0.934	0.933
F1 score, F score, F measure	0.941	0.927	0.933	0.934
P value (compared with BiLSTM)	—	—	—	< 0.001
P value (compared with Transformer)	—	—	—	< 0.005

From the data in Table 3, the model performs relatively balanced across the three states, with no significant deficiency in any single category. Among them, the recognition sensitivity for the baseline state reaches the highest value of 0.958, which benefits from the stability of physiological signals under calm conditions. In the more complex task of stress recognition, the model still maintains a sensitivity of 0.915 and a specificity of 0.971, demonstrating its robustness in distinguishing high-arousal negative emotions. The high UAR of 0.935 indicates that the model effectively addresses the class imbalance issue. Paired t-tests against the baseline models show that the performance improvements of GABT-Net over BiLSTM and the standard Transformer are both statistically significant ($p < 0.001$ and $p < 0.005$, respectively), confirming that these improvements are not caused by random errors.

To further verify the superiority of GABT-Net on the WESAD dataset, this study selects three representative state-of-the-art (SOTA) works published in 2024 for horizontal comparison. These include the negative emotion recognition model based on CNN-LSTM proposed by Sun et al. [31], the edge computing model based on two-dimensional feature maps [32], and the multimodal deep learning framework proposed by Abd Al Aleem et al. [33]. The comparison results are presented in Table 4.

Table. 4 Performance comparison between WESAD dataset and other methods

Method	Usage mode	Network structure	Accuracy (%)	P value of accuracy compared with GABT-Net	F1 score (%)	P value of F1 score compared with GABT-Net
Sun et al. (2024) [31]	ECG, EDA, EMG (Electromyogram)	CNN-LSTM	93.84	0.004	91.20	0.003
Sun et al. (2024) [32]	BVP (Blood Volume Pulse), EDA, Temp	Two-dimensional CNN (edge computing)	94.50	0.012	92.15	0.009
Abd Al Aleem et al. (2024) [33]	ECG, EDA, RESP (Respiration)	Hybrid deep learning	95.12	0.021	93.80	0.035
GABT-Net	BVP, EDA, Temp, ACC (Accelerometer)	GAT +Bayesian Transformer	96.20	-	94.31	-

The comparative data in Table 4 shows that GABT-Net outperforms existing state-of-the-art (SOTA) methods in both accuracy and F1-score. Compared with the CNN-LSTM architecture adopted by Sun et al., the accuracy of this model is improved by 2.36%. Even when compared with the well-performed hybrid deep learning model proposed by Abd Al Aleem et al., GABT-Net still maintains an accuracy advantage of approximately 1%.

The p-values compared with GABT-Net are obtained by paired t-test and are all significant at the 0.05 level. GABT-Net is significantly superior to existing state-of-the-art methods in both accuracy and F1-score. This performance superiority is mainly attributed to the explicit modeling of implicit collaborative relationships among physiological signals by the graph attention network (GAT) structure, rather than merely conducting fusion at the feature level. In addition, the uncertainty calibration mechanism introduced by the Bayesian framework endows the model with stronger anti-interference ability when processing physiological signals with high noise, thereby improving the boundary clarity of overall classification. The calculation efficiency of each method is analyzed as shown in Table 5:

Table. 5 Comparison of calculation efficiency of each model

Method	Parameters (M)	FLOPs (G)	Inference delay (ms, batch=1)	Peak memory (MB)
Sun et al.	1.2	2.1	25.7	979
Sun et al.	1.0	1.8	18.5	852
Abd Al Aleem et al.	1.5	2.6	28.3	1024
GABT-Net	0.85	1.24	12.4	486

As Table 5 shows, GABT-Net has advantages in computational efficiency. Specifically, its parameter count is only 0.85M, which is reduced by about 29%~43% compared with other models, indicating that the model structure is lighter and easier to deploy on devices with limited computing resources. The floating-point operations (FLOPs) are 1.24G, much lower than the other three methods, which further reflects its high efficiency in real-time processing of multi-modal physiological signals. In terms of inference latency, GABT-Net takes 12.4 ms for single-sample prediction, which is reduced by about 31%~56% compared with other models, and can meet the real-time prediction requirements under high-frequency physiological signal acquisition. In addition, the peak memory usage is only 486 MB, nearly half or even lower than that of other methods, showing the feasibility of running in edge devices or embedded systems. Based on the above indicators, it shows that GABT-Net achieves lower computational cost and faster inference speed while maintaining or exceeding the classification performance of existing models. Compared with traditional CNN-LSTM or hybrid deep learning models, GABT-Net extracts multi-modal collaboration features through GAT and performs Bayesian time series prediction. It enhances performance, and effectively compresses model size and computational overhead, achieving a balance between performance and efficiency.

Considering the feasibility of practical model deployment and the potential impact of data processing strategies on results, this study conducts evaluations from two dimensions: computational resource consumption and different data splitting strategies. Table 6 presents the hardware-aware performance metrics on the NVIDIA RTX 4090 platform, as well as the prediction accuracy under different sliding window step sizes.

Table. 6 Computational efficiency analysis and robustness evaluation under different sliding window steps

Category	Indicators/conditions	Numerical value/performance
Computational efficiency	Parameter quantity	0.85 M
	Floating point operand	1.24 G
	Inference delay (Batch = 1)	12.4 ms
	Peak memory usage	486 MB
Robustness (accuracy)	Step size = 1 second (high overlap)	96.20%
	Step size = 10 seconds (partial overlap)	95.45%
	Step size = 60 seconds (no overlap)	94.12%

The data in Table 6 demonstrate that GABT-Net maintains a reasonable computational load while preserving high performance. With a parameter count of 0.85M and 1.24G floating-point operations, it has the potential for deployment on edge devices, and the inference latency of 12.4 milliseconds fully meets the requirements of real-time monitoring. In the robustness test of sliding window strategies, as the step size increased from 1 second to 60 seconds (i.e., complete non-overlapping), the accuracy of the model exhibited an expected slight decline (from 96.20% to 94.12%) but did not experience a catastrophic drop. Even under the non-overlapping setting with zero risk of data leakage, its performance remains competitive. This result confirms that the model has learned the essential features of physiological states rather than merely memorizing redundant information brought by overlapping windows.

To verify the rationality of the fully connected graph construction strategy and analyze the internal learning mechanism of the model, this study compares different graph construction methods and conducted a visualization analysis of the learned attention weights. Table 7 shows the impact of fully connected graphs, k-nearest neighbor graphs, and sparse graphs based on correlation thresholds on model performance.

Table. 7 Influence of different graph construction strategies on model performance

Graph construction strategy	Accuracy (%)	Macro average F1(%)
Fully connected graph (the method in this paper)	96.20	94.31

K nearest neighbor graph ($k = 3$)	93.15	90.85
Correlation threshold graph (> 0.5)	94.02	91.70
Random graph	88.45	85.20

The experimental results clearly demonstrate that the fully connected graph strategy is significantly superior to sparse graph construction methods. The fully connected structure achieves the highest accuracy of 96.20%, which verifies that in physiological monitoring scenarios, there exist potential global dependencies among all collected physiological indicators (e.g., heart rate, electrodermal activity, body temperature). Any artificial sparsification operation may lead to the loss of critical synergistic information.

A further in-depth analysis of the attention weight distribution in the fully connected graph reveals that the model successfully captures connection patterns with physiological significance. In the stress state, the average attention weight connecting the electrodermal activity node and the heart rate variability node increases by approximately 40% compared with that in the baseline state. This dynamic weight redistribution mechanism is highly consistent with the regulatory rules of the autonomic nervous system: sympathetic nerve activation can synchronously induce increased sweat gland secretion and altered heart rhythm. By means of the GAT layer, GABT-Net adaptively enhances information aggregation between these two types of key indicators, which proves that it is statistically effective, and possesses the interpretability to capture the real physiological linkage mechanism.

To further verify the contribution of the core modules of GABT-Net to model performance, an ablation experiment is designed to evaluate the effects of the GAT collaborative modeling module and the Bayesian time series prediction module, respectively. The specific settings are as follows: Model A removes the GAT module and only feeds the original feature sequence into the Bayesian Transformer. Model B retains the GAT module but replaces the Bayesian Transformer with a standard Transformer (deterministic prediction). Model C is the complete GABT-Net, which retains both the GAT and Bayesian Transformer modules. The experimental results are shown in Table 8:

Table. 8 Ablation experimental result

Model variant	Graph Attention (GAT)	Bayesian Transformer	Accuracy (%)	Macro F1 (%)
A	×	√	92.87	90.12
B	√	× (standard Transformer)	94.56	92.03
C	√	√	96.20	94.31

Table 8 suggests that each module contributes significantly to model performance. After removing the GAT module, the model accuracy drops to 92.87% and Macro F1 decreases to 90.12%, indicating that the collaboration relationship among multi-modal physiological signals is crucial for mental state recognition. After removing the Bayesian time series prediction module, the accuracy is 94.56% and Macro F1 is 92.03%, showing that probabilistic prediction and uncertainty quantification can significantly improve temporal modeling capability and classification reliability. The complete GABT-Net achieves the highest values in both Accuracy and Macro F1, further verifying the complementarity of the two core modules, GAT and Bayesian Transformer, both of which are indispensable. Overall, the ablation experiment fully demonstrates the rationality of the model design. The GAT module can capture the instantaneous collaboration features of different physiological indicators, while the Bayesian Transformer module can quantify temporal dependence and prediction uncertainty. The combination of the two modules achieves optimal classification performance and reliability and provides a theoretical basis for the subsequent deployment of the model in practical mental state monitoring.

4.4 Discussion

The experimental results strongly demonstrate the superiority of the GABT-Net framework, whose success stems from two core innovations. First, the GAT encoder explicitly models multimodal physiological signals as a relational graph, providing a more comprehensive state representation. This is fundamentally different from current mainstream fusion strategies. For example, although the advanced fusion models proposed by Wang et al. (2025) [34], Liang (2024) [35], and Xu et al. (2024) [36] perform well in their respective fields, they focus on integration at the feature level. In contrast, our method implements "relational fusion" within the physiological domain, prioritizing the capture of the collaborative network structure between signals, thereby achieving significant performance improvement. Second, the introduction of the Bayesian framework directly addresses the key issue of model reliability. The extremely low ECE and the strong correlation between prediction entropy and accuracy prove that GABT-Net is accurate and a trustworthy model that can understand its own knowledge limitations. These findings hold important theoretical and practical significance. Theoretically, this study verifies a novel "structure-temporal" modeling paradigm, promoting research toward the development of AI systems with intrinsic calibration and uncertainty awareness. Practically, it provides more accurate physical and mental state analysis for athletes and offers a solid technical blueprint for developing more intelligent and safer daily health wearable devices.

5. Conclusion

5.1 Research Contribution

This study successfully addresses two major challenges in current physiological signal analysis: the lack of collaborative modeling and the unreliability of prediction results. It proposes and verifies a novel GABT-Net framework. First, the framework uses GAT to effectively capture the instantaneous collaborative relationships among multimodal physiological signals. Then, it employs Bayesian Transformer to perform probabilistic prediction on the dynamic evolution of state sequences. Experimental results on the public WESAD dataset show that the model achieves a final average accuracy of up to 96.2%, which is significantly better than all baseline models. More importantly, the contribution of this study lies in providing a new "structure-temporal" modeling paradigm with both high accuracy and high reliability (realized through uncertainty quantification). The study lays a solid theoretical and technical foundation for building more intelligent and credible psychological-physiological state monitoring systems.

5.2 Future Works and Research Limitations

Despite the positive outcomes of this study, there are still limitations that point to directions for future research. First, this study uses the WESAD dataset collected in a laboratory environment, and the applicability of its results in real-world, dynamic sports scenarios needs further verification. Second, although the current model demonstrates good generalization ability, it has not yet been deeply personalized for specific individuals. Third, the model learns the correlation between physiological indicators rather than the underlying causal mechanisms. Future work will focus on three aspects: 1. Applying the model to real "in-the-wild" datasets of athletes. 2. Exploring fast personalization methods based on transfer learning or meta-learning. 3. Attempting to integrate causal discovery techniques to enhance the interpretability and guiding value of the model, and ultimately developing a real-time feedback system prototype.

Funding: This research received no external funding.

Data Availability Statement

The datasets used and/or analyzed during the current study are available from the corresponding author Hesong Zhao on reasonable request via e-mail zhaohs2024@163.com.

Competing Interest

The authors declare no competing financial or non-financial interests.

Author Contributions Declaration

Yuantao Cheng: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation

Hesong Zhao: writing—review and editing, visualization, supervision, project administration, funding acquisition

Ting Gong: methodology, software, validation, formal analysis, investigation

ETHICS STATEMENT

The study was conducted in accordance with the Declaration of Helsinki, the studies involving human participants were reviewed and approved by Department of Physical Education, Chengdu Textile College Ethics Committee (Approval Number: 2023.10399003). The participants provided their written informed consent to participate in this study. All methods were performed in accordance with relevant guidelines and regulations.

References

- [1] Haque, Y., Zawad, R. S., Rony, C. S. A., Al Banna, H., Ghosh, T., Kaiser, M. S., & Mahmud, M. (2024). State-of-the-art of stress prediction from heart rate variability using artificial intelligence. *Cognitive Computation*, 16(2), 455-481.
- [2] Van Hooren, B., Mennen, B., Gronwald, T., Bongers, B. C., & Rogers, B. (2025). Correlation properties of heart rate variability to assess the first ventilatory threshold and fatigue in runners. *Journal of sports*

- sciences, 43(2), 125-134.
- [3] Liang, Z., Geng, G., Song, Q., & Tang, M. (2024). A time-synchronized multimodal monitoring system for general anesthesia. *Medicine in Novel Technology and Devices*, 23, 100324.
 - [4] Sakamoto, S. I., Hutabarat, Y., Owaki, D., & Hayashibe, M. (2023). Ground reaction force and moment estimation through EMG sensing using long short-term memory network during posture coordination. *Cyborg and Bionic Systems*, 4, 0016.
 - [5] Morgenroth, J., Kalenchuk, K., Moreau-Verlaan, L., Perras, M. A., & Khan, U. T. (2023). A novel long-short term memory network approach for stress model updating for excavations in high stress environments. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*, 17(1), 196-216.
 - [6] Cai, Y., Li, X., & Li, J. (2023). Emotion recognition using different sensors, emotion models, methods and datasets: A comprehensive review. *Sensors*, 23(5), 2455.
 - [7] Sweeney-Fanelli, T. C., & Imtiaz, M. H. (2024). Ecg-based automated emotion recognition using temporal convolution neural networks. *IEEE Sensors Journal*, 24(18), 29039 - 29046.
 - [8] Wang, Y., Tian, W., Xu, J., Tian, Y., Xu, C., Ma, B., ... & Liu, H. (2023). Wearable wireless dual-channel EEG system for emotion recognition based on machine learning. *IEEE Sensors Journal*, 23(18), 21767-21775.
 - [9] Zong, X., Zhang, C., Zhang, N., Wang, Z., & Wang, J. (2025). Breathable, superhydrophobic and multifunctional Janus nanofibers for dual-mode passive thermal management/facial expression recognition with deep learning. *Chemical Engineering Journal*, 505, 159759.
 - [10] Ahmed, A., Ramesh, J., Ganguly, S., Aburukba, R., Sagahyroon, A., & Aloul, F. (2023). Evaluating multimodal wearable sensors for quantifying affective states and depression with neural networks. *IEEE Sensors Journal*, 23(19), 22788-22802.
 - [11] Nandini, D., Yadav, J., Singh, V., Mohan, V., & Agarwal, S. (2025). An ensemble deep learning framework for emotion recognition through wearable devices multi-modal physiological signals. *Scientific Reports*, 15(1), 17263.
 - [12] Bhatti, A., Behinaein, B., Hungler, P., & Etemad, A. (2024). Attx: Attentive cross-connections for fusion of wearable signals in emotion recognition. *ACM Transactions on Computing for Healthcare*, 5(3), 1-24.
 - [13] Shen, Y. (2025). TransformHR: a novel approach to athlete heart rate monitoring with transformer-based wearable technology. *Internet Technology Letters*, 8(6), e631.
 - [14] Toğaçar, Ş., Tel, M., & Cömert, Z. (2025). Fatigue detection in performance sports with multivariate time series analysis: integrated use of wavelet transform and transformer models. *Signal, Image and Video Processing*, 19(13), 1147.
 - [15] Phan, D. T., Choi, J., Vo, T. T., Ngo, D., Lee, B. I., & Oh, J. (2024). Multi-Sensor Wearable Device With Transformer-Powered Two-Stream Fusion Model for Real-Time Leg Workout Monitoring. *IEEE Journal of Biomedical and Health Informatics*, 29(4), 2534 - 2545.
 - [16] Sharifi-Renani, M., Mahoor, M. H., & Clary, C. W. (2023). BioMAT: An open-source biomechanics multi-activity transformer for joint kinematic predictions using wearable sensors. *Sensors*, 23(13), 5778.
 - [17] Liang, F., Ou, L., Lei, Z., Tu, X., & Xin, K. (2025). Spatial-Temporal Transformer for Optimizing Human Health Through Skeleton-Based Body Sports Action Recognition. *IEEE Access*, 13, 112476 - 112493.
 - [18] Hou, Y., Zhu, Q., Lai, Z., Zhong, W., Yu, Q., Wang, L., ... & Zhong, Y. (2025). SensoriMind-Trans Net: EEG and sensorimotor-driven transformer for athlete potential evaluation. *Frontiers in Psychology*, 16, 1496013.
 - [19] Yu, F., Yu, C., Tian, Z., Liu, X., Cao, J., Liu, L., ... & Jiang, M. (2024). Intelligent wearable system with motion and emotion recognition based on digital twin technology. *IEEE internet of things journal*, 11(15), 26314-26328.
 - [20] Wang, J., Qiao, L., Lv, H., & Lv, Z. (2022). Deep transfer learning-based multi-modal digital twins for enhancement and diagnostic analysis of brain mri image. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 20(4), 2407-2419.
 - [21] Moser, M. K., Ehrhart, M., & Resch, B. (2024). An explainable deep learning approach for stress detection in wearable sensor measurements. *Sensors*, 24(16), 5085.
 - [22] Li, J., & Washington, P. (2024). A comparison of personalized and generalized approaches to

- emotion recognition using consumer wearable devices: Machine learning study. *JMIR AI*, 3(1), e52171.
- [23] Zhu, J., Deng, F., Zhao, J., Liu, D., & Chen, J. (2023). Uaed: Unsupervised abnormal emotion detection network based on wearable mobile device. *IEEE Transactions on Network Science and Engineering*, 10(6), 3682-3696.
- [24] Liu, Y., Zavareh, A. T., & Zoghi, B. (2024). Enhancing Well-Being and Alleviating Stress via Wearable-Driven Emotion Recognition and EQ Intentional Practice with Deep Reinforcement Learning. *IEEE Sensors Letters*, 9(1), 6000504.
- [25] Yang, L., Amin, O., & Shihada, B. (2024). Intelligent wearable systems: Opportunities and challenges in health and sports. *ACM Computing Surveys*, 56(7), 1-42.
- [26] Wu, J., Mo, Z., Gao, X., Xin, W., Shi, W., & Park, J. (2025). Artificial intelligence assisted wearable flexible sensors for sports: research progress in technology integration and application. *International Journal of Smart and Nano Materials*, 16(3), 510-548.
- [27] Mai, N. D., Nguyen, H. T., & Chung, W. Y. (2024). Deep learning-based wearable ear-ecg emotion recognition system with superlets-based signal-to-image conversion framework. *IEEE Sensors Journal*, 24(7), 11946-11958.
- [28] Zhu, J., Ye, Z., Ren, M., & Ma, G. (2024). Transformative skeletal motion analysis: optimization of exercise training and injury prevention through graph neural networks. *Frontiers in neuroscience*, 18, 1353257.
- [29] Dubey, P., Dubey, P., Zakariah, M., Almazyad, A. S., & Alsekait, D. M. (2025). TRANSHEALTH: A Transformer-BDI Hybrid Framework for Real-Time Psychological Distress Detection in Ambient Healthcare. *Computers, Materials and Continua*, 85(2), 3897-3919.
- [30] Kegel, J. L., Kazman, J. B., Clifton, D. R., Emanuele, P., Nelson, D. A., & Deuster, P. A. (2023). The combined effects of coping and pain interference on army readiness. *Frontiers in Pain Research*, 4, 1175574.
- [31] Sun, J., Portilla, J., & Otero, A. (2024, December). Negative emotion recognition based on physiological signals using a CNN-LSTM model. In *2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)* (pp. 3736-3741). IEEE.
- [32] Sun, J., Portilla, J., & Otero, A. (2024). A deep learning approach for fear recognition on the edge based on two-dimensional feature maps. *IEEE journal of biomedical and health informatics*, 28(7), 3973-3984.
- [33] Abd Al Aleem, M., Mubarak, R., Salem, N. M., & Sadek, I. (2024, October). A Deep Learning Approach Using WESAD Data for Multi-Class Classification with Wearable Sensors. In *2024 6th Novel Intelligent and Leading Emerging Sciences Conference (NILES)* (pp. 194-197). IEEE.
- [34] Wang, P., Wang, H., & Zhang, H. (2025). Cognitive Workload Assessment in Aerospace Scenarios: A Cross-Modal Transformer Framework for Multimodal Physiological Signal Fusion. *Multimodal Technologies and Interaction*, 9(9), 89.
- [35] Liang, S. (2024). Enhancing Multimodal Emotional Information Extraction in Film and Television through Adaptive Feature Fusion with DenseNet, Transformer, and 3D CNN Models. *Applied Artificial Intelligence*, 38(1), 2419609.
- [36] Xu, M., Cui, J., Ma, X., Zou, Z., Xin, Z., & Bilal, M. (2024). Image enhancement with art design: a visual feature approach with a CNN-transformer fusion model. *PeerJ Computer Science*, 10, e2417.